Day 6: Introduction to Network Devices Theory :Overview of switches, routers, hubs, firewalls, and access points. Practical :Identify ports and functions of network devices in a setup. Task: Create a table comparing network devices and their features .

Tasks:-

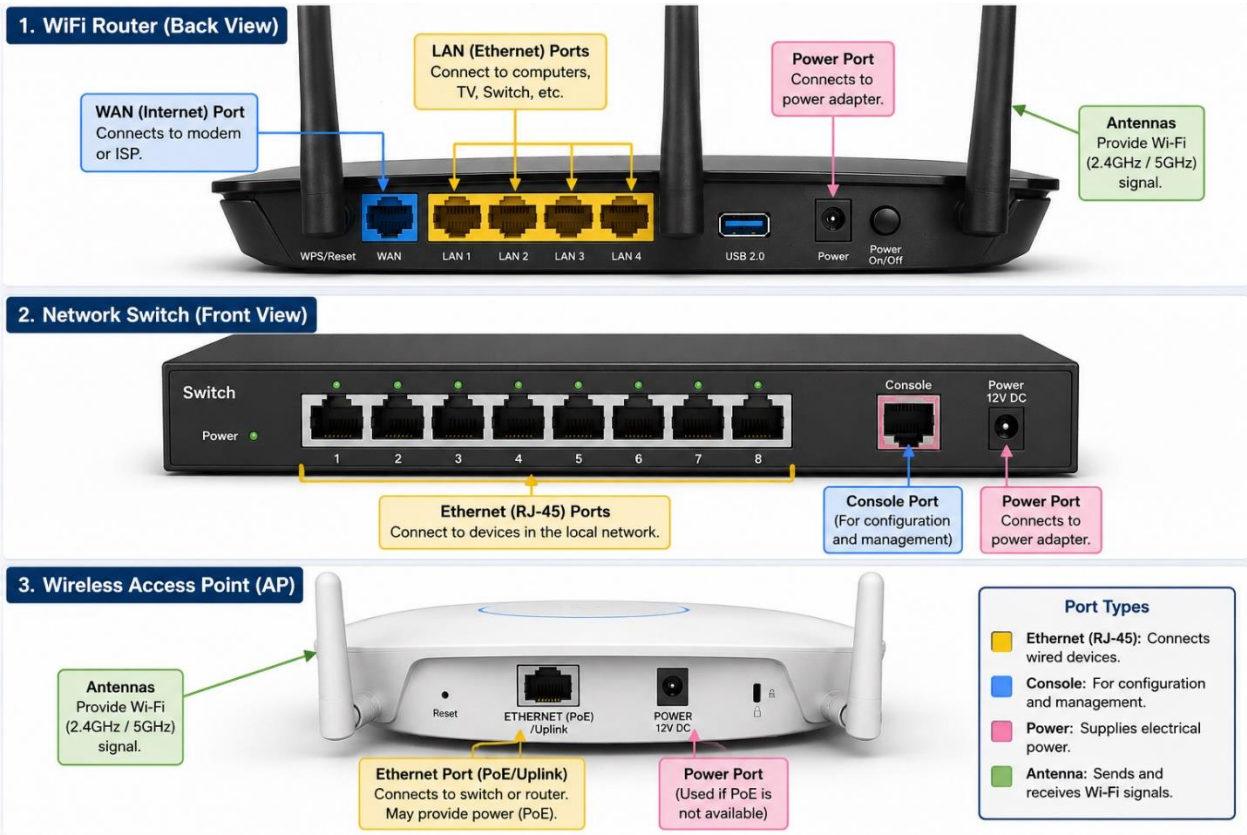
1. Create a table listing the main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home WiFi setup.

Ans.:-

Network Device	Main Function	Typical Ports / Interfaces
Switch	Connects multiple wired devices within the same local network and forwards data to the correct device using MAC addresses.	5–48 Ethernet (RJ-45) LAN ports, 1–4 SFP/SFP+ ports (on managed switches), Console port (on enterprise models)
Router	Connects the home network to the Internet and routes data between different networks. Often provides DHCP, NAT, and firewall features.	1 WAN Ethernet port, 2–8 LAN Ethernet (RJ-45) ports, USB port (optional), Power port
Hub	Connects multiple devices and broadcasts incoming data to all connected devices without filtering.	4–24 Ethernet (RJ-45) ports, Power port
Firewall	Monitors and filters incoming and outgoing network traffic to protect the network from unauthorized access and cyber threats.	1 WAN port, 2–8 LAN Ethernet ports, Console/Management port (on dedicated devices), USB port (optional)
Access Point (AP)	Provides wireless (Wi-Fi) connectivity for devices and extends or creates a wireless network.	1 Ethernet (RJ-45) uplink port (often PoE), Power port (or PoE), Reset button, Wireless (2.4 GHz/5 GHz/6 GHz) interfaces

2. Given images or diagrams of network devices, label each device and mark the ports (Ethernet, console, power, antenna) for at least three devices.

Ans.:-



3. Pick a popular app you use daily (like Instagram, Zomato, or Paytm) and describe in 3-4 lines how at least two different network devices (from switches, routers, firewalls, access points) play a role when you use that app.

Ans.:-

When I open Instagram, my smartphone first connects to the Wi-Fi Access Point, which provides a stable wireless connection to my home network.

The Router assigns an IP address using DHCP and forwards my requests (such as loading the feed, uploading photos, or sending messages) to Instagram's servers over the internet.

Switches in my ISP's network and Instagram's data centers quickly direct data packets between servers so posts, reels, and messages reach me without delay.

A Firewall continuously monitors and filters network traffic, blocking unauthorized access and malicious requests to help keep my account, personal information, and connection secure.

**4. Suppose your college WiFi is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason for the issue and how upgrading one device could help.

Hint: Think about device capacity, coverage, and traffic handling.**

Ans.:-

Possible Reason:-

The college WiFi may be slow and disconnect frequently because the Wi-Fi Access Point (AP) is overloaded. If too many students connect to the same AP at the same time, it cannot handle all the traffic efficiently, leading to slow speeds, weak signals, and frequent disconnections.

Suggested Upgrade:-

Upgrading to a modern Wi-Fi 6 (802.11ax) Access Point or installing additional access points in busy areas would improve performance. A Wi-Fi 6 AP can support more users simultaneously, provide better coverage, reduce network congestion, and deliver faster and more stable internet connections. This would result in smoother online classes, faster downloads, and fewer WiFi interruptions for students.